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Cypher Analytica — R Programming for Data Analysis and Statistical Computing

Level: Beginner to Intermediate, with introductory statistical modeling
Duration: 60 instructor-led hours + 30 hours of independent practice
Suitable for: Students, researchers, aspiring data analysts, business analysts, and working professionals
Prerequisites: Basic computer knowledge. Familiarity with descriptive statistics is helpful; no prior programming experience is required.

Core tools: R, RStudio, tidyverse, ggplot2, and Quarto
Teaching approach: Concept explanation, live coding, dataset exercises, statistical interpretation, and a complete analysis project.

Course objectives

By the end of the course, learners should be able to:

- Write and debug R scripts.
- Import, clean, transform, and combine datasets.
- Explore data using numerical summaries and visualizations.
- Select and apply introductory statistical tests.
- Build and evaluate basic regression models.
- Produce reproducible reports with clear interpretations.

Module-wise curriculum

Module	Topics covered	Hours
1. Introduction to R and RStudio	Applications of R; installation and environment setup; console, scripts, and Help; RStudio projects; working directories and relative paths; installing and loading packages; naming conventions; reproducible workflows	3
2. R Fundamentals and Data Structures	Objects and assignment; numeric, integer, character, and logical types; vectors, matrices, arrays, lists, and data frames; tibbles; factors; indexing and subsetting; type conversion; vectorized operations; NA, NaN, and NULL	6

Module	Topics covered	Hours
3. Programming Logic and Functions	Operators; if, else, and ifelse; for and while; writing functions; arguments and return values; scope basics; apply-family introduction; introductory purrr::map; warnings, errors, debugging, and tryCatch	5
4. Data Import, Export, and Inspection	Reading CSV, delimited text, and Excel files; importing RDS files; inspecting dimensions, structure, and types; selecting sheets and ranges; managing parsing problems; exporting cleaned data; database connectivity concepts	4
5. Data Cleaning and Transformation	dplyr functions: select, filter, arrange, mutate, group_by, and summarise; pipes; missing values; duplicates; recoding; case_when; string handling with stringr; date handling with lubridate; identifying outliers and documenting treatment decisions	7
6. Reshaping and Combining Data	Tidy data principles; pivot_longer and pivot_wider; separating and combining columns; row and column binding; joins; key uniqueness; unmatched records; many-to-many join risks; validating row counts and totals	4
7. Exploratory Data Analysis	Frequency tables; proportions; mean, median, and mode; quantiles; variance and standard deviation; grouped summaries; distributions; cross-tabulations; covariance and correlation; distinguishing association from causation	5
8. Data Visualization with ggplot2	Grammar of graphics; aesthetic mappings; geometries; bar charts, histograms, density plots, box plots, scatterplots, and line charts; facets; scales; labels; themes; missing-data handling; accessible colors; exporting figures	7
9. Statistical Inference and Hypothesis Testing	Sampling concepts; confidence intervals; hypotheses; p-values; effect sizes; assumptions; independent and paired t-tests; chi-square tests; one-way ANOVA; introductory nonparametric alternatives; multiple-testing awareness	6

Module	Topics covered	Hours
10. Regression and Introductory Modeling	Simple and multiple linear regression; categorical predictors; coefficient interpretation; residual diagnostics; multicollinearity concepts; introductory logistic regression; training and test data; leakage prevention; regression and classification evaluation basics	5
11. Reproducible Reporting	Organizing scripts; documenting analysis decisions; using set.seed; recording package information; Quarto documents; combining narrative, code, tables, and charts; rendering an HTML report	3
12. Capstone Project and Assessment	Final data checks; analysis validation; report refinement; presentation of findings and limitations; practical assessment and viva	5
Total	**Guided instruction, practical work, and assessment**	**60**

Practical exercises

Learning area	Suggested task
Data structures	Create and subset a student marks dataset
Functions	Write reusable functions for summaries and input checks
Data import	Import an employee workbook and resolve parsing issues
Data cleaning	Standardize categories, dates, and missing-value codes
Transformation	Summarize sales by month, region, and product
Reshaping and joins	Convert wide survey responses to long format and attach participant details
Exploratory analysis	Examine salary distributions and department-level differences
Visualization	Create a set of charts answering specific business questions

Learning area	Suggested task
Hypothesis testing	Compare independent groups and report assumptions, effect sizes, and uncertainty
Regression	Model a continuous outcome and interpret coefficients and diagnostics
Reporting	Produce a reproducible HTML report with code, results, and interpretations

Capstone project options

1. **Retail Sales Analysis:** Clean transaction data, analyze product and regional performance, visualize trends, and explain business findings.
2. **Employee Analytics:** Examine salary, tenure, and department patterns using summaries, visualizations, and an appropriate statistical comparison.
3. **Student Performance Analysis:** Explore marks, attendance, and study-related variables; build a basic regression model and discuss its limitations.
4. **Survey and Research Data Analysis:** Clean questionnaire responses, apply documented scoring rules, summarize results, and conduct justified statistical tests.
5. **Customer Behavior Analysis:** Examine purchase frequency, order value, customer groups, and repeat-purchase patterns.

Students should develop their project progressively during the course, using the final module for validation and presentation.

Mandatory project deliverables

- A clearly stated analysis question and data dictionary.
- Organized scripts with reproducible data preparation.
- Documented treatment of missing values, duplicates, and unusual observations.
- At least five purposeful visualizations.
- An appropriate statistical test or model, with justification and assumption checks.
- Interpretations covering practical meaning, uncertainty, and limitations.

- A rendered HTML report and its editable source.
- A README explaining how to reproduce the analysis.

Suggested assessment structure

Component	Weight
Programming and data-cleaning assignments	25%
Exploratory analysis and visualization assessment	20%
Statistical analysis and interpretation	20%
Capstone project	25%
Technical viva and reproducibility demonstration	10%
Total	**100%**

Certification Criteria: Minimum 60% overall, completion of mandatory assignments, and successful demonstration of a reproducible capstone analysis.

Interview and research preparation

Practice explaining vectors versus lists, factors versus character variables, missing-value handling, vectorization, grouped calculations, join duplication, statistical assumptions, p-values versus effect sizes, and regression interpretation.